

Ziad Hazem

Bachelor of Science

Computer & Systems Department, Faculty of Engineering Helwan University

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GitHub Profile

LinkedIn Profile

EDUCATION

• Faculty of Engineering, Helwan University

Sep. 2019 - May 2024

Bachelor of Science (B.Sc.) in Computer Engineering

Cumulative: 88.97%

Graduation Project: **Excellent**

Class Rank: **First** in Computer Engineering Department

EXPERIENCE

• Software Engineer

March - August 2024

Xenio.co

Remote

- Interacted with various AWS services such as EC2 and S3, contributing to the overall cloud infrastructure management.
- Developed and adapted an OpenAI-based chatbot to function as a human-like customer service agent with minimal costs, utilizing prompt engineering and solid software engineering principles.
- Integrated the chatbot with WhatsApp Business, enhancing customer interaction capabilities.
- Improved the overall system design and coding practices, ensuring high-quality, maintainable code.
- Followed agile methodologies and managed tasks efficiently using Jira.
- **Technologies & Professional Skills:** Node.js, Express, TypeScript, OpenAI, AWS SDK, Meta API, Webhooks, Cloud Computing, Software Development, Prompt Engineering, System Design, Agile Methodologies, Task Management, and Testing.

• Freelance Software Engineer

May 2022 - Feb 2024

Upwork

Remote

- Achieved Top Rated status, placing in the **top 10% of freelancers** on the platform.
- Worked on various projects for companies like **Parmonic Ltd** and **Mehr Capital (deal.ai)** using the MERN stack (React.js, Next.js, Express, MongoDB, and Node.js).
- Delivered high-quality software solutions while collaborating with diverse teams and clients.

• Research Developer

Oct 2020 - May 2022

Wireless Research Lab (WRL)

Helwan University

- Created and developed an easy-to-use e-commerce web system for Helwan Nanotechnology Center using OOP, React.js, Next.js, TypeScript, CKEditor, and more.
- Contributed to a research project funded by **AAST** for a Laboratory Learning Management System, focusing on the backend with Node.js, Express, and TypeScript.
- Integrated the backend with an existing codebase in .NET Core and Myrtle, ensuring seamless functionality and performance.

PERSONAL PROJECTS

• Lin-K-it

A hands-on learning solution combining an embedded lab kit, a user-friendly web platform, and an immersive desktop application.

- Addresses high student volume, shortage of teaching assistants, and financial barriers to hands-on learning experiences.
- Provides a tactile experience that merges the realism of physical experimentation with the benefits of technology.
- Features an AI-powered auto-grading tool for instant feedback and an AI-driven assistant for teachers to manage lab sessions efficiently.
- Technology Used: Tauri, React.js, Python, FastAPI, Rust, Embedded C, RAG, OpenAI

• Contributed to Evilginx

Custom version of Evilginx utilizing a browser-based reverse proxy for improved stealth.

- Modified Evilginx to use a browser-based reverse proxy instead of a traditional one, enhancing evasion from service providers.
- Implemented a notification and alert system to improve user experience and responsiveness.
- Technology Used: Go.

• Zaki The Assistant

An advanced, personalized intelligent chatbot leveraging ChatGPT capabilities.

- Created a user-friendly standard ChatGPT foundation for ongoing development.
- Implemented a feature to extract user data from 'pdf/doc/csv/txt' files for custom bot training.
- Addressed limitations, storage handling, and abuse prevention for user packages.
- Technology Used: OpenAI, Node.js, TypeScript, AWS-SDK, NLP.

TECHNICAL SKILLS AND INTERESTS

Languages: JavaScript, TypeScript, Python, C/C++, Rust
Libraries: React.js, Python Libraries, D3.js, Puppeteer
Web Development Tools: Node.js, Express, FastAPI, Git, GitHub, Docker
Frameworks: React.js, Next.js
Cloud/Databases: MongoDB, Firebase, AWS (EC2, S3), MySQL
Message Brokers: RabbitMQ, MQTT, Kafka
Architectural Styles: Microservices
ORMs: Sequelize, Mongoose
Relevant Skills: OpenAI, Prompt Engineering, JWT, JWT Security
Relevant Coursework: Data Structures & Algorithms, Operating Systems, Object-Oriented Programming, Database Management Systems, Software Engineering
Methodologies: Agile Methodologies, SOLID Principles, Jira
Areas of Interest: Web Design and Development, Cloud Security
Soft Skills: Problem Solving, Self-learning, Presentation, Adaptability

VOLUNTEERING

- **Academic Head** *Sep 2022 - July 2023*
Pixels Egypt **Helwan University**
 - * Assumed the role of Academic Head, demonstrating strong leadership and organizational skills.
 - * Oversaw various Computer Science courses, including Web Development and Embedded Systems.
 - * Instructed Embedded Systems programming and Security courses using AVR, including an introduction to ARM architecture.
 - * Led a dedicated team in the development of a Nursing Arm featuring innovative bloodless blood glucose measurement techniques.
- **Academic Instructor** *Oct 2021 - Sep 2022*
Pixels Egypt **Helwan University**
 - * Instructed Embedded Systems programming and Security courses using AVR, including an introduction to ARM architecture.
 - * Assisted in the development of various educational materials and resources for students.